

## TOPIC 5: APPROXIMATION

### 1. Meaning of Approximation

Approximation means finding a value that is close to the exact value but easier to use. We approximate to a given number of decimal places, significant figures, or to the nearest ten, hundred, etc.

### 2. Approximation to Decimal Places

Look at the digit after the required place. If it is 5 or more, round up; if less than 5, leave the digit as it is.

Example: 3.14159 to 2 d.p. = 3.14; to 3 d.p. = 3.142.

### 3. Approximation to Significant Figures

Significant figures start from the first non-zero digit. Example: 0.00456 to 2 s.f. = 0.0046; 34567 to 3 s.f. = 34600.

### 4. Approximation to Nearest Tens, Hundreds, Thousands, etc.

Example: 347 to nearest ten = 350; 347 to nearest hundred = 300; 4,560 to nearest thousand = 5,000.

### 5. Tenths, Hundredths, Thousandths

Same as 1 d.p., 2 d.p., 3 d.p. respectively.

### 6. Application

Used in measurement, money, estimation and quantitative reasoning when an exact value is not needed.

### 7. Summary

Approximation simplifies numbers while remaining close to the true value. Rules for rounding depend on the next digit (5 or more → round up).

End of Topic 5